

HUMBERSIDE, United KingdomWMO: **033735**Lat: **53.574N**Lon: **0.351W**Elev: **121**StdP: **14.63**Time Zone: **0.00 (EUW)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.5	29.9	22.9	17.3	30.5	25.2	19.3	32.4	33.0	51.5	30.6	49.5	6.3	210	0.650

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	78.8	65.9	75.3	64.2	73.1	62.7	67.4	75.2	65.6	72.9	64.1	70.7	11.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	91.4	70.7	62.8	86.0	68.5	61.2	81.0	66.8	31.8	75.3	30.5	73.0	29.4	71.1	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.5	25.3	22.7	21.2	85.5	7.5	3.2	15.8	87.8	11.4	89.7	7.2	91.5	1.7	93.9		
(5)			20.5	71.2	7.3	1.7	15.3	72.5	11.0	73.5	6.9	74.4	1.6	75.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.0	40.3	41.2	44.8	48.7	53.4	58.5	63.1	62.7	58.3	53.2	45.8	41.7		
	DBStd	9.53	5.23	5.81	5.13	5.09	5.12	4.54	4.18	4.16	4.29	4.90	5.68	6.71		
	HDD50	1273	303	252	176	82	25	1	0	0	0	22	148	264		
	HDD65	5166	767	668	628	488	360	199	87	96	205	369	577	722		
	CDD50	1648	1	5	13	44	130	257	407	393	250	120	21	7		
	CDD65	68	0	0	0	0	1	6	29	25	5	2	0	0		
	CDH74	397	0	0	0	4	12	50	193	99	30	9	0	0		
	CDH80	47	0	0	0	0	1	8	25	10	2	1	0	0		
(14) Wind	WSAvg	11.3	12.4	12.4	11.9	10.9	10.7	9.5	9.4	10.7	10.7	11.8	12.0	12.9		
(15) Precipitation	PrecAvg	25.6	2.0	1.8	1.7	1.9	1.8	2.4	2.3	2.1	2.5	2.4	2.5	2.4		
	PrecMax	33.6	3.5	3.0	3.6	5.4	3.7	9.2	4.4	4.6	5.0	4.4	5.0	4.2		
	PrecMin	15.9	0.4	0.3	0.4	0.4	0.4	0.4	0.9	0.2	0.7	0.6	1.7	1.2		
	PrecStd	4.4	0.9	0.7	0.8	1.3	0.9	2.0	1.1	1.1	1.2	0.9	1.0	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.8	59.1	64.1	70.0	75.4	81.0	83.8	81.0	77.4	71.7	62.4	57.5		
		MCWB	52.9	53.0	53.8	58.0	64.5	68.6	68.5	66.7	65.9	63.4	59.1	54.3		
	2%	DB	53.4	55.2	59.2	64.7	70.2	73.8	79.1	76.9	73.1	66.2	58.9	55.5		
		MCWB	50.4	51.1	52.3	55.2	60.5	63.2	66.5	65.4	63.0	61.2	55.8	52.7		
	5%	DB	50.3	52.1	55.7	61.2	66.6	71.3	75.4	73.7	69.5	62.8	55.8	53.6		
		MCWB	47.9	48.5	50.3	53.7	57.8	61.8	64.2	62.7	61.9	58.8	53.2	51.2		
	10%	DB	48.2	50.0	53.4	58.7	64.1	68.1	73.2	71.6	66.3	60.8	53.8	51.5		
		MCWB	46.5	47.4	48.4	52.2	56.0	60.4	63.0	61.8	59.6	56.9	51.3	49.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.0	53.8	55.2	60.0	65.0	69.7	70.2	68.3	67.2	63.8	59.1	55.3		
		MCDB	56.9	57.3	60.2	68.4	74.0	78.8	80.6	76.1	74.1	70.6	61.8	57.2		
	2%	WB	50.9	51.6	53.0	56.5	61.9	65.5	67.6	66.7	64.6	61.7	56.3	53.3		
		MCDB	52.9	54.7	58.3	62.6	69.0	71.9	75.8	74.3	71.0	65.7	58.6	55.7		
	5%	WB	48.5	49.6	51.1	54.8	59.4	63.1	65.8	65.0	62.4	59.6	54.1	51.7		
		MCDB	49.9	51.9	55.4	60.6	65.4	69.0	73.1	72.1	67.4	62.7	56.4	53.7		
	10%	WB	46.3	47.5	49.3	52.5	57.1	61.0	64.2	63.5	60.7	57.4	52.1	49.6		
		MCDB	47.7	49.6	52.9	57.8	62.1	66.8	70.7	69.4	65.2	60.3	54.1	51.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	10.3	13.3	14.9	15.2	15.0	16.2	15.7	14.5	11.7	10.0	9.1		
		MCDBR	11.3	12.0	17.2	20.4	20.7	20.2	21.0	18.5	18.5	13.5	10.7	10.3		
		MCWBR	10.1	8.8	11.5	12.8	12.0	11.7	10.5	8.9	10.5	9.3	9.2	9.3		
	5% WB	MCDBR	11.2	11.7	17.0	19.4	18.9	18.0	19.0	15.8	15.6	12.6	10.6	10.2		
		MCWBR	10.4	9.2	12.0	12.7	12.2	11.6	11.0	8.7	10.0	9.3	9.4	9.4		
(40) Clear-Sky Solar Irradiance	taub		0.324	0.341	0.372	0.398	0.397	0.404	0.406	0.394	0.372	0.353	0.326	0.311		
	taud		2.350	2.388	2.301	2.244	2.294	2.297	2.304	2.358	2.420	2.448	2.403	2.325		
	Ebn at Noon		202	240	255	263	270	268	266	262	253	233	201	182		
	Edh at Noon		18	24	32	39	39	39	38	34	29	23	17	16		
(44) All-Sky Solar Radiation	RadAvg		208	423	772	1202	1514	1562	1520	1258	921	522	257	165		
	RadStd		21	50	89	140	102	140	151	82	83	43	27	17		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page